

Placing Practice and Expertise at the Center of Explainable AI: A Participatory Design Approach to Explainable AI for Enterprise Information Architects

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Abstract

AI presents significant opportunities to address long-standing challenges in Information Architecture (IA) practices within enterprises. However, research shows that shortcomings in the transparency and understandability of AI are hindering trust and limiting the adoption of effective solutions for Information Architecture. To overcome these barriers, this research places practice and experience at the center of Explainable AI (XAI) by collaborating with Information Management (IM) practitioners to develop an industry-based, context-aware approach to XAI based on their expertise. The paper outlines the participatory co-design development process implemented for the *Explainable AI-to-EIM Lifecycle (XEL)* workshop, enhancing the usefulness of AI applications for Enterprise Information Architecture. The research preliminary findings demonstrate how AI capabilities can be translated to meet practitioners' needs, and by aligning the workshop's activities with high-level design aspects for XAI, the paper presents a human-centered XAI workshop design framework that is reusable, translatable, and transferable to other industry-based XAI contexts.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Empirical studies in HCI**;

Keywords

Explainable AI, Human-Centered XAI, Participatory Design Research, Enterprise Information Management, Enterprise Information Architecture

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1 Introduction

Information Management (IM) practitioners struggle with long-standing issues in managing vital business information that are rooted in the challenges of complex Enterprise Information Architectures (EIA), including information silos, poor information findability, difficulties in sharing information, and concerns about information security [14., 28., 36.]. In the context of Enterprise Information Management (EIM), information architecture serves as a blueprint for managing information assets across the organization and encompasses a wide range of practices and procedures, including information governance policies, information management processes and the building of metadata and taxonomies (including controlled vocabularies) for information assets. EIA enables IM practitioners to organize, describe, manage, and control vital information in support of organizational goals and for compliance and assurance purposes [18., 36.]. Despite being a well-established concept within information systems, the implementation of EIA often falls short of achieving the intended goal of facilitating more effective IM [14., 28., 36.], with research showing that many organizations continue to lack sound information governance [28.]. Legacy and poorly integrated document management systems are still widely used, and there is inconsistent use of metadata and taxonomy [36.]. Metadata is labor intensive, often managed manually, with concerns about information quality, findability, control, and permissions [7.], highlighting a critical need for IM practitioners to explore the use of advanced AI tools for automating and facilitating better EIA in practice.

However, current research shows that information managers are sceptical about the use of AI solutions in their work and raise concerns about its transparency, trustworthiness, and suitability for maintaining vital business information [22., 28., 36.]. There is a burgeoning need for new approaches to Explainable AI (XAI) that will make intelligent systems more accessible and understandable to practitioners [1., 8., 15.]. The AI community has made significant progress in developing algorithmic approaches to XAI, by improving the interpretability of complex AI models and providing greater accountability and explainability through AI-generated outputs that validate software operations [1., 15.]. However, employing one opaque software solution to interpret another offers little value to practitioners tasked with explaining the benefits of AI for information governance in large organizational settings, such as private businesses, public hospitals, or government departments [16.]. In practice, trust in AI will not come solely from the operation of a self-interpreting, self-validating system, *explainability* in this study

is seen as a human-centered capability developed by practitioners to understand, interpret, and make AI useful in their work. Theoretically this research proposes re-centring AI explainability through a Human-Computer Interaction (HCI) lens that prioritizes the human perspective in designing XAI [8., 9., 25.]. Extending this XAI view requires that practitioners are not seen merely as passive recipients or end-users in a requirements elicitation exercise, but that their expertise is brought to creating context-aware, industry-based XAI approaches [4., 6., 9.].

To implement this, our research team is collaborating with a leading Australian Enterprise Information Management consultancy to develop an Explainable AI workshop through a participatory co-design process that first aims to bring the industry awareness of experienced IM consultants to the design of a workshop that is human-centered, context-based and built from their expertise. In this case focusing on how AI can meet the information architecture needs of practitioners across the EIM lifecycle, from acquisition to the disposition of essential business information. We first developed the workshop by concentrating on developing a series of problem-solving activities that could be stepped through to aligning existing AI applications with industry's most recognized EIM challenges. We next took the *Explainable AI-to-EIM Lifecycle (XEL)* workshop out into practice, inviting key information managers from three large organizations to join us in collaborative roundtable workshops where they shared their expertise as industry experts, offering insights into their experiences with AI and their critical EIM needs. The workshop was successful in allowing the practitioners to develop an understanding of how AI could work to meet the needs they identified as most critical. Moreover, by considering the steps we developed for the XEL workshop more broadly, with high-level design aspects for XAI at large, this paper introduces an XAI workshop design framework that is reusable, translatable, and transferable, enabling its adaptation to other Explainable AI industry contexts.

2 Related Work

2.1 Positioning Enterprise Information Architecture in Enterprise Information Management Context

While various interpretations of information architecture exist in the literature [2., 18.], this paper positions EIA, in the context of EIM, as the structured design and organization of information assets across an organization to ensure they are accessible, usable, and aligned with organizational goals. If EIM is the set of disciplines, processes and practices that organizations adopt to structure, manage, describe, and govern information assets for their goals and objectives [5., 14., 19.], then *the architecture* is the blueprint for presenting how this is instantiated in each organization. While EIA is shaped by all aspects of information management, it tends to center on the design and description of enterprise information assets, with the development and use of taxonomies (controlled vocabularies) and metadata schemas critical for describing resources to meet the strategic demands for information use. While the importance of EIA is well-established in research [17., 18., 33.], industry observations reveal that EIA is rarely understood holistically as part of a strategic EIM framework, and with the manual effort required to

develop and maintain taxonomies and metadata schemas for sound resource description often under-resourced, EIM often comes up short for achieving organizational goals and meeting regulatory compliance and assurance requirements [14., 28., 36.].

2.2 AI Applications for Enterprise Information Architecture in Practice

Advances in AI, particularly the increased availability and use of generative AI (Gen-AI) and Large Language Models (LLMs), offers an important opportunity to redress many of the long-standing EIM challenges associated with EIA practices. By enabling better contextual understanding through natural language processing, semantic reasoning or dynamic content generation, AI can improve current EIA practices, like taxonomy development and metadata curation permanently, with significant improvements in adaptability, scalability, and interoperability for information assets [13., 30., 37.]. To make an information asset easier to understand, locate, and use, taxonomy and metadata work hand in hand, where metadata provides descriptive information about the contextual, structural and administrative attributes of an information asset, while taxonomy ensures consistent terms (controlled vocabularies) are applied to those metadata attributes across the organization [13., 30.]. Existing research demonstrates how Gen-AI can transform approaches to metadata curation and taxonomy development [13., 20., 30., 37.]. For instance, Sundaram et al. [30.] investigated the role of LLMs in supporting metadata curation, standardization, and correction to improve adherence to metadata standards with structured domain knowledge. Zhang et al. [37.] underscored the importance of incorporating context-based information into LLMs for subject inference in metadata curation and annotation tasks. Gunn et al. [13.] have explored the use of generative AI for innovative and scalable taxonomy creation. With AI features available to transform information architecture practices and address EIM challenges, a critical question arises: How understandable are these AI technologies to IM practitioners in their everyday work environments, where they must explain their application, use, and value to the organization before putting them into use?

2.3 Participatory Design for Human-Centered Explainable AI

With the widespread integration of AI into work practice, bringing a *human-centered* perspective to XAI is essential [9., 16., 25.]. As stated by Ehsan et al. [10.] "Understanding *who* opens the black-box of AI is just as important as opening it", to better understand XAI stakeholders' needs and objectives, recent research has categorized XAI stakeholders into different groups, including expertise-oriented groups (based on levels of Machine Learning expertise) [23., 35.], role-oriented groups (focused on functional roles associated with AI systems) [21., 32.], and knowledge-based groups (decomposing expertise into knowledge and contexts) [31.]. Despite useful insights into XAI needs within each category, characteristics of different stakeholder groups (e.g. perceptions, training, experience, and demographic characteristics) can impact their XAI needs and therefore the XAI methods to be developed [8., 16., 21., 24.]. For instance, Ehsan et al. [8.] evaluated the impact of people's AI experience with various types of AI explanations, while Kim et al.

[16.] investigated the impact of varying domain knowledge and AI experience on people's responses to XAI explanations. Nimmo et al. [24.] investigated the impact of characteristics, such as age, gender, experience, and personality traits, on people's trust and understanding of AI systems. Although these studies aimed to explore how XAI can achieve better understanding with stakeholder groups, the practitioners who will use the system are often treated as passive recipients or evaluators and are not actively involved in developing the XAI process itself. With less attention being paid to the context in which different industry practitioners will work with AI systems when developing XAI approaches, there is currently a shortcoming in addressing the context-based needs of practitioners [29.]. This is particularly true for IM practitioners, who primarily interact with AI at the application level and often have no control over the AI tools they can use in their work environment [36.]. This research joins others in proposing to better align with practitioners' context-specific needs in the development of XAI approaches by applying Participatory and Collaborative Design (participatory co-design) methodologies, joining both researchers and practitioners in the design process [10., 38.]. A participatory co-design approach to XAI will improve the representation of practitioners' context-specific needs [26.] and should enhance ownership of the XAI approach and system use [11.]. Moreover, participatory co-design allows for the use of flexible and creative methods to elicit tacit knowledge, fostering mutual learning among all participants [10., 11., 26.].

3 Participatory Design WORKSHOP Study

To incorporate industry awareness into the development of an XAI workshop and address the actual needs of IM practitioners, our participatory co-design commenced by partnering with IM consultants from a leading Australian Information Management consulting firm. With extensive experience analyzing clients' EIM needs and developing Information Architectures to achieve their clients' information governance objectives, the IM consultants possess a deep understanding of IM "pain points". Additionally, they are well-equipped to provide context- and language-fit checks for workshop activity design, making them ideal collaborators in the first stage of workshop design. The consultants contributed valuable insights and informed the design of activities in the XEL workshop through two industry briefing sessions, followed by their participation in a mock workshop, undertaken to review activities and assist in language use and context-based design, including developing supplementary materials for use in the XEL workshop.

3.1 XAI-to-EIM Lifecycle (XEL) Workshop Overview

The XEL workshop is designed around hands-on activities to prompt active engagement and participation and, unique to our approach, roundtable problem-solving that allows participants to work collaboratively, share insights, and propose solutions from their positions as IM industry experts. While the XEL workshop is designed for face-to-face participation in the room, it is facilitated through the use of Miro [39.] - a virtual collaboration whiteboard tool, allowing all participants to simultaneously access and interact with the activities from their laptops with actions and responses projected to the screen in real-time to facilitate discussion with

the opportunity to examine choices, acknowledge alternatives and adjust. As illustrated in Figure 1 the XEL workshop consists of an icebreaker activity and two main activities to investigate how AI can better meet organizations' critical IM needs and support information architecture practices. Roundtable discussion points were developed to prompt the participants to bring their expertise to challenges in making AI applications in EIM more transparent and trustworthy.

3.1.1 Activity 1: AI Experiences in Enterprise Information Management (Miro projected to screen, plus Post-it Notes and Whiteboard).

Activity 1(a) begins by engaging participants in charting AI capabilities across the EIM Lifecycle. In this activity, IM practitioners are provided with labels identifying current AI applications and perceived opportunities in EIM and placing these across various stages of the EIM Lifecycle, from acquisition to disposition [19.]. To balance engagement and participation without turning this activity into a guessing and writing exercise, labels derived from an Environmental Scan of AI currently being integrated into EIM platforms in the vendor market [36.], representing common AI capabilities for EIM tasks are provided. These labels, which participants can click and drag simultaneously to align with EIM lifecycle stages, include an "undefined" option for participants to add their own inputs (an example from the Miro is shown in Figure 2). After identifying AI use across the EIM Lifecycle, participants are asked to evaluate their satisfaction with their AI experiences. Instead of using a detailed Likert scale, we deliberately introduced a simple three-point satisfaction scale represented by emojis: "Above expectation 😊" - indicating that the new AI approach has improved the current approach, "Too soon to know 😐" - indicating that the new AI approaches have not yet been fully evaluated, and "Below expectation ☹️" - indicating that the new AI approach did not improve the current approach. Our objective is not to establish a statistical measure for the participants; rather, the aim of this scale is to stimulate open discussions about the responses among them. Following this activity, we opened a roundtable discussion with "What key challenges or concerns do you have about using AI" to engage participants prior to commencing Activity 1(b).

Activity 1(b) focuses on IM practitioners' challenges and concerns with AI use in the EIM context. This activity is designed as a physical whiteboard exercise, incorporating tangible elements and changing the pace of engagement through a varied mode of interaction that fostered a more immersive and collaborative experience for participants by moving from the Miro Board pairs into the room [27.]. Participants write their responses on Post-it notes and stick them to a whiteboard, where the researcher subsequently works with the participants to organize and group the notes into similar themes. Figure 3 illustrates IM practitioners performing Activity 1(b) in practice. Roundtable discussion resolves the issue, "What is the biggest concern about using AI? How do you think we can tackle it?" asks our IM practitioners to tackle a roundtable challenge directly and share their proposed solutions as industry experts.

Activity 1(c) builds upon IM practitioners' identified AI uses for specific EIM tasks to explore various AI application development and deployment pathways prioritized by IM practitioners. Participants are provided with three predefined AI development pathways

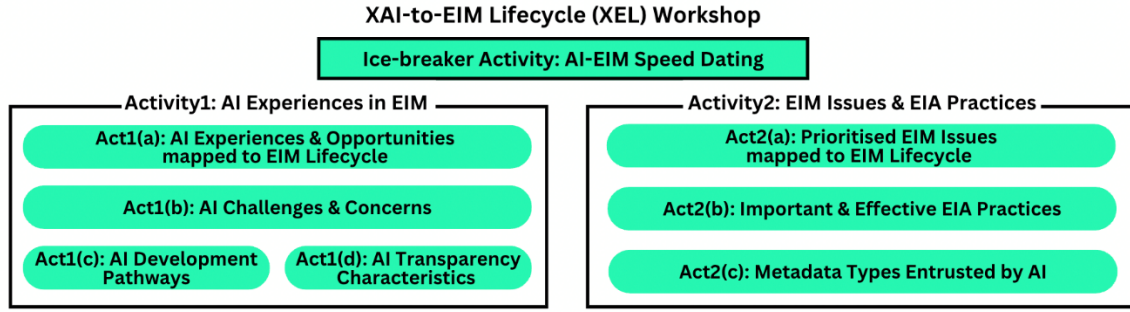


Figure 1: XAI-to-EIM Lifecycle (XEL) Workshop Activity Design Overview

Pick Up AI Use / AI Opportunities OR Define your own (click+drag)

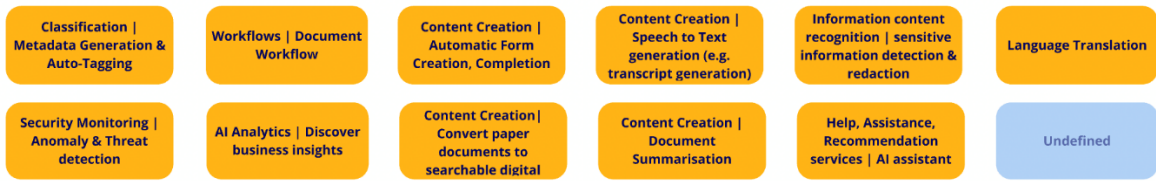


Figure 2: Example of Activity Labels on Miro Board (Activity 1(a))

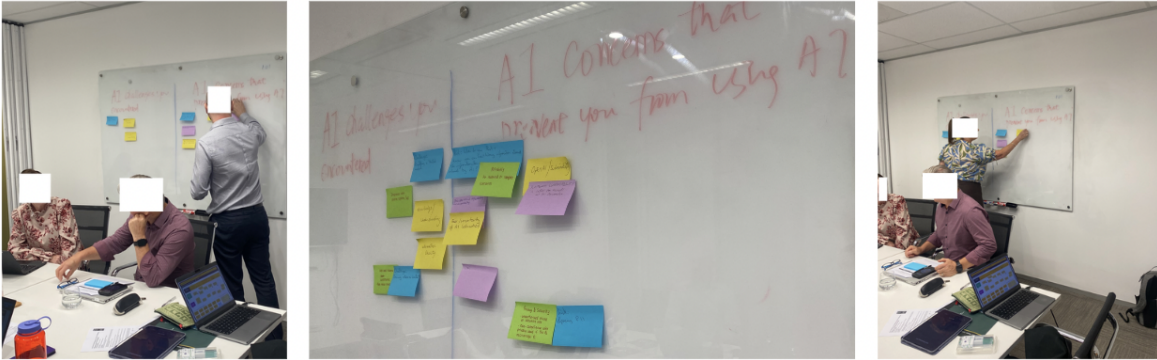


Figure 3: IM Practitioners perform Activity1(b) on Whiteboard in practice

derived from prior research [36.], along with an “Other AI development pathway” (i.e. undefined) option to allow for additional input, and asked to order their preferences. Following this activity, the roundtable question, “What is the biggest challenge you consider for the chosen AI development pathway(s)?” brings participants’ expertise and insights to a direct discussion of the AI development considerations and constraints faced by organizations.

With the issues and concerns regarding AI applications in EIM identified, Activity 1(d) prompts IM practitioners to identify the AI transparency characteristics they consider most important in the EIM context. Common AI transparency types are crafted in appropriate IM language as labels for participants to select, with an “Undefined” option available for participants to provide additional input. Roundtable discussion addressed the challenge “For your industry overall, how can AI transparency lead to better AI

understanding and adoption?” aiming to bring participants’ insights to requirements and strategies needed to achieve greater AI transparency in EIM.

3.1.2 Activity 2: Enterprise Information Management Issues and Enterprise Information Architecture Practices (Miro projected to screen). Activity 2 then transitions to identifying critical EIM needs to explore how AI can support EIA practices. Activity 2(a) begins with participants prioritizing IM issues associated with IA across the EIM Lifecycle. Labels representing common IM issues, derived from prior research on EIM issues [28., 36.] are provided, along with an option (e.g. “Undefined”) for participants to contribute their own inputs. Following this, IM practitioners addressed the roundtable question: “What is the biggest IM problem, and how can AI help to tackle it?”, where participants share their experiences and approaches with possible solutions drawn from the AI

insights generated in the previous activities. Activity 2(b) then engaged participants in positioning their perceived IA practices (e.g., using taxonomy or metadata schemas, or applying information governance policies and procedures) onto both the *Important* and *Effective IA Practice Bullseye* charts to evaluate the current state of IA practice in their organizations. Predefined IA practice labels, derived from prior research [18., 36.], were provided, with the option for participants to define their own labels. Activity 2(c) continues by examining AI trust for supporting EIA practice at the structural level of information design, where participants identify the types of metadata that can be fully entrusted to an AI approach, versus those where human intervention is required to manage critical information assets. Following Activities 2(b) and 2(c), the IM roundtable addressed the challenge: “What needs to happen to increase trust for using AI to manage your most critical information?”

3.2 XAI-to-EIM Lifecycle (XEL) Workshop Implementation

The XEL workshop was undertaken across two rounds in November 2024, by inviting nine IM professionals representing three large Australian organizations in different sectors (e.g. Utility, Mining and Manufacturing), who also had some familiarity with AI in their daily practice (although prior AI experience was not required) to participate. Each workshop lasted approximately four hours, including a lunch break, and provided participants with supplementary materials describing the labels and terms used in the activities. A third round is scheduled for early March 2025, and all participants will join a post-workshop evaluation session scheduled for April 2025 for member-checking [12., 38.].

3.2.1 Data Collection. Due to the blended activity and discussion format of the workshop design, various streams of data were collected. The workshops were audio-recorded and transcribed, while Miro activity responses were exported directly in CSV format for analysis. Additionally, the Post-it notes created during the physical whiteboard activities were collected and digitized, along with screenshots of the whiteboard.

3.2.2 Data Analysis. To analyze the data collected from the XEL workshops, the study applies thematic analysis, primarily following an inductive, bottom-up approach to develop categories and themes. While the analysis focuses on the transcripts of the recorded roundtable discussions to identify important themes, the research also emphasizes the value of solutions the roundtable members (participants) propose to specific industry challenges for XAI posed by the questions asked, these are evaluated to establish collective industry insight relevant to major issues, and further thematic analysis is applied to these responses. Activity data, including Miro CSV files, Post-it notes and researcher observations are integrated into the thematic analysis [3., 34.]. At this stage, both the workshop series and the evaluation session are ongoing, while preliminary analysis has taken place and informs findings reported in this paper, completing the analysis of all data from the workshops will not take place until after the final stages, including a further industry workshop and an evaluation session that will involve all participants. Final analysis of the workshop, and industry evaluation of the workshop design, will focus on assessing the

suitability of the XAI roadmap presented as a useful tool that can be made available for independent use by IM practitioners. At the next stage, we hope to make the overarching framework presented here, also freely available for wider use.

4 Preliminary Findings

In this section, we present preliminary outcomes from the XEL workshop including results showing how making AI capabilities explainable in context can address the prioritized IM issues identified by IM practitioners across the EIM Lifecycle.

4.1 Aligning AI Capabilities with Critical EIM Issues Across the EIM Lifecycle

Putting AI tools and solutions into IM practice requires making them more useful, with greater explainability. To achieve this, it is essential to empower the people who will put AI to work. IM practitioners are struggling to trust AI because of a lack of transparency and understandability: firstly, without transparency practitioners frequently lack a holistic understanding of the current AI capabilities available to them, and secondly, they find it challenging to directly understand and explain to their organizations how these AI capabilities link to the organizational pain points they aim to resolve. By working with predefined AI capability labels and using the EIM Lifecycle as a shared lingua franca in industry, the XEL workshop successfully enabled IM practitioners to work through these concerns and develop solutions for aligning AI capabilities with their prioritized issues. A chart of AI capabilities aligned with prioritized IM issues, based on preliminary analysis from the first two workshops is summarized in Table 1, showing recurring IM issues like Poor Metadata & Taxonomy Management across multiple EIM stages and suggesting that the same AI capability can address issues across different stages. Conversely, different AI capabilities were also identified as applicable to addressing the same IM issues. Our research proposes that the AI-IM issues road mapping practitioners undertake in the XEL workshop can serve IM practitioners in adopting AI solutions by allowing them to develop a context-based XAI approach to their critical IM needs. In subsequent work, once data from the third round of the XEL workshop is collected, all data will be integrated for analysis to incorporate the rich insights from IM practitioners’ roundtable proposals. For now, we have highlighted three key points from the early roundtable analysis: 1) A practice-based use guide for AI solutions addressing specific IM tasks is needed as a significant part of AI explainability for “time-poor” IM practitioners. 2) The “sustainability” (i.e., how long an AI tool will remain in the market) of specific AI tool-based solutions is essential for AI explainability (and transparency) in the workplace. 3) Transparency in AI data use, including AI training data, input data used by AI systems, and AI-generated data consumed by practitioners raises significant security and access control concerns for business-critical information.

4.2 XAI Workshop Design Scheme in Practice-based Contexts

By considering the steps developed for the XEL workshop more broadly, this paper contributes to the HCI community by establishing an XAI workshop design framework with seven high-level

Table 1: Practitioner charting AI to Prioritized IM Issues across the EIM Lifecycle

EIM Lifecycle	AI Capability	Prioritized IM Issues
Acquisition & Creation	Classification Metadata Generation & Auto-Tagging	Poor Taxonomy & Metadata Management
Organization & Description	Content Creation Convert Paper to Digital Searchable	Paper-driven Processes
	Classification Metadata Generation & Auto-Tagging	Poor Taxonomy & Metadata Management
		Poor Information Quality
Search & Retrieval	AI Analytics Content Analytics	Information Overload & Silos
		Poor Information Findability
	Classification Metadata Generation & Auto-Tagging	Information Overload & Silos
		Poor Information Quality
Sharing & Collaboration	Security Monitoring Anomaly & Threats Detection	Poor Taxonomy & Metadata Management
		Vital Information Security
	Classification Metadata Generation & Auto-Tagging	Poor Internal & External Sharing
		Poor Access Controls
Utilization & Decision-Making	Information Content Recognition Sensitive Information Detection & Redaction	Information Overload & Silos
	AI Analytics Discover Business Insights	Legal & Compliance
Disposition & Reuse	Classification Metadata Generation & Auto-Tagging	Value Creation & Knowledge Discovery
		Applying AI & Advanced Technologies
		Retention & Disposal
		Legal & Compliance

Table 2: Reusable XAI Workshop Design Scheme for Practice-based Contexts

Key Design Aspects	Description
Domain-specific AI awareness	Develop domain-specific awareness and embed in the design of activities and resources used in XAI workshop design, e.g. This study undertook an Environmental Scan of AI integration in EIM platforms to prepare common AI capabilities labels used in XEL Workshop.
Context-based understanding of industry medium	Provide a medium (<i>lingua franca</i>) to connect AI capabilities to challenges faced in work-based contexts, e.g. This study applied the EIM Lifecycle as the medium shared by IM practitioners to connect AI capabilities to prioritized IM Issues.
Participatory XAI workshop design	Take a participatory co-design approach, involving industry practitioners in XAI workshop design <i>prior to implementation</i> , e.g. This study conducted briefing sessions followed by a mock XEL workshop with IM consultants to ensure a co-design solution prior to implementation.
Prioritize and include participants' expertise	Participants are experts and understand demands for explainability in their fields, providing valuable insights and recommendations. Rather than being treated as end-users, they engage in in-depth problem-solving by sharing experiences with AI in the workplace and generating richer, more meaningful outcomes, e.g. For this study, each activity in the XEL workshop is followed by a problem for resolution, seeking IM practitioners' deliberation and insight for the topic.
Activity and problem-based workshop design	Design the XAI workshop with activities that address industry prioritized problems faced by practitioners, ensuring engaged participation. Developing XAI should not be seen as a requirement gathering, or a focus group interview session, e.g. In this study, the XEL workshop is activity based with a problem-driven approach that requires engagement.
Provision of workshop supporting document	Prepare supporting documentation on AI-related concepts to be explored in the workshop to help participants with diverse backgrounds & AI knowledge levels navigate and engage in the workshop activities effectively, e.g. In this study, we provide supporting documentation.
Cross-sector collaboration of participants	Participants from different sectors tend to be more open to discussion and sharing their thoughts on the topic at hand, when there is no direct competition participants collaborate to reach mutually beneficial outcomes, e.g. In this study, we invited participants from different sectors and then paired them together in the icebreaker activity.

design aspects for XAI, that is reusable, translatable, and transferable, enabling its adaptation to other XAI industry contexts. Table

2 provides details for each design aspect, along with descriptions and references to the XEL workshop development process.

5 Conclusion

This paper addresses current shortcomings in developing Explainable AI (XAI) approaches to meet industry needs, through a project that demonstrates how the real needs of information management (IM) practitioners can be aligned with AI solutions across the enterprise information management (EIM) lifecycle as it is understood and applied in their work practices. By addressing the need to place practitioners' experience and expertise at the center of XAI, this research shows how collaboration with IM practitioners resulted in a human-centered, industry-based XAI workshop and points to the need to continue to prioritize participatory co-design process that leverages practitioner expertise in realizing XAI. The preliminary outcomes of the XEL workshop demonstrate how making AI capabilities explainable *in context* was effective in addressing the prioritized IM issues identified by practitioners across the EIM lifecycle, enhancing the usefulness of AI applications for their needs in enterprise information architecture. Additionally, this research contributes to the HCI community by establishing a high-level design framework for XAI workshops that is reusable, translatable, and transferable, enabling its adaptation to other XAI industry contexts.

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